INCH-POUND
MIL-PRF-3098/129C
May 17, 2004
SUPERSEDING
MIL-PRF-3098/129B
1 October 1997

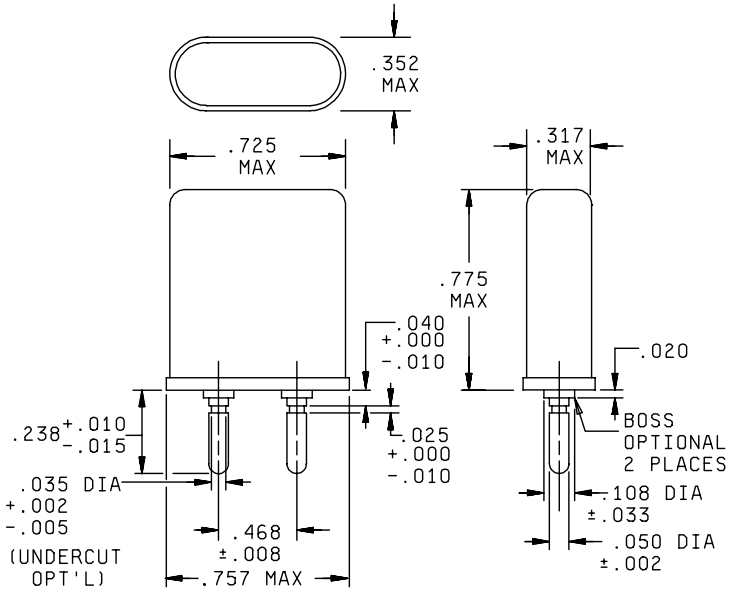
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR148/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 10 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	.05	.020	.51	.050	1.27	.486	12.34
.005	.13	.025	.64	.108	2.74	.725	18.42
.008	.20	.030	.76	.238	6.05	.757	19.23
.010	.25	.035	.89	.317	8.05	.775	19.69
.015	.38	.040	1.02	.352	8.94		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR148/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10.0 MHz to 61.0 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Mode of oscillation: Third mechanical overtone.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Equivalent resistance: 51 ohms.

Resistor voltage drop: 0.32 volt.

Resonance: Series.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Frequency tolerance: ±0.0005 percent.

Equivalent resistance: 40 ohms, maximum.

Unwanted modes: Minimum unwanted-to-main mode resistance ratio of 2:1.

Shock:

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Temperature cycling:

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±0.0005 percent.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5955-0760)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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